

KRASNY CHIKOY, Russian Federation

WMO: 309350

Lat: **50.367N**Lon: **108.750E**Elev: **771**StdP: **92.40**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.4	-35.2	-39.7	0.1	-37.2	-37.4	0.1	-35.2	5.4	-15.0	4.7	-15.4	0.1	70	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	30.1	18.0	28.0	17.0	26.0	16.3	19.5	26.6	18.5	24.9	17.6	23.5	2.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	13.6	21.6	16.3	12.8	20.9	15.5	12.1	20.1	59.3	26.5	56.0	24.9	53.0	23.6	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.5	4.5	-40.0	34.6	2.4	2.2	-41.8	36.1	-43.2	37.4	-44.6	38.6	-46.3	40.2		
(5)			-39.7	21.6	2.5	1.5	-41.5	22.6	-43.0	23.5	-44.4	24.3	-46.3	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.9	-23.9	-18.4	-7.7	2.4	9.3	16.0	18.9	16.3	9.1	-0.1	-12.1	-21.3
	DBStd	15.60	5.90	6.01	6.53	4.34	4.28	3.66	2.81	3.26	4.02	4.62	6.39	6.68
	HDD10.0	4702	1050	796	549	229	66	2	0	1	63	313	662	971
	HDD18.3	7083	1308	1029	807	477	281	87	26	79	276	572	912	1229
	CDD10.0	737	0	0	0	2	43	183	277	196	37	0	0	0
	CDD18.3	77	0	0	0	0	1	17	44	16	1	0	0	0
	CDH23.3	1154	0	0	0	2	83	357	508	187	17	0	0	0
	CDH26.7	342	0	0	0	0	19	105	169	48	2	0	0	0
Wind	WSAvg	1.4	0.8	1.0	1.6	2.2	2.3	1.8	1.5	1.4	1.4	1.3	1.1	0.9
Precipitation	PrecAvg	340	5	3	3	9	26	52	88	89	39	11	7	7
	PrecMax	515	11	7	10	24	74	106	186	181	71	34	26	19
	PrecMin	244	1	0	0	1	3	11	35	37	13	1	1	2
	PrecStd	66	2	2	3	7	16	25	38	33	15	8	5	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.3	-0.1	12.0	21.5	28.9	32.5	33.7	30.7	25.6	17.2	4.5	-3.5
		MCWB	-6.4	-3.3	5.1	9.9	14.3	17.9	20.6	19.0	15.5	9.0	0.6	-5.1
	2%	DB	-9.3	-3.2	7.9	17.6	25.6	29.4	30.6	27.6	22.1	13.5	1.3	-6.4
		MCWB	-10.5	-5.5	2.5	7.7	12.9	16.7	18.8	17.7	13.3	6.3	-1.4	-7.6
	5%	DB	-12.3	-6.0	4.9	14.4	22.7	27.3	28.4	25.2	19.3	10.4	-0.8	-8.8
		MCWB	-13.2	-7.7	0.3	6.0	11.4	16.0	18.1	17.1	12.1	4.6	-2.9	-9.8
	10%	DB	-14.6	-8.5	2.1	11.5	19.7	25.2	26.1	22.9	16.5	7.5	-2.8	-11.0
		MCWB	-15.3	-9.9	-1.4	4.5	9.9	15.2	17.3	16.3	10.7	3.1	-4.4	-11.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.5	-3.2	5.2	10.3	15.1	19.1	21.5	20.4	16.2	9.0	0.9	-5.0
		MCDB	-5.3	-0.4	11.3	20.7	26.4	28.5	31.0	27.6	22.5	16.7	4.0	-3.6
	2%	WB	-10.6	-5.5	2.6	8.1	13.5	17.9	20.0	18.9	14.5	6.5	-1.2	-7.6
		MCDB	-9.4	-3.3	7.7	16.4	24.4	26.4	27.7	24.9	20.2	12.6	1.0	-6.5
	5%	WB	-13.2	-7.6	0.5	6.3	11.9	16.9	19.1	17.9	13.0	5.0	-2.8	-9.7
		MCDB	-12.4	-6.0	4.4	13.6	21.1	24.9	25.6	23.1	18.0	9.9	-0.9	-8.9
	10%	WB	-15.3	-9.9	-1.2	4.8	10.4	16.0	18.2	17.0	11.5	3.3	-4.4	-11.7
		MCDB	-14.7	-8.6	2.0	11.3	18.8	23.1	24.1	21.7	15.7	7.0	-2.8	-10.9
Mean Daily Temperature Range	5% DB	MDBR	11.4	15.4	15.1	14.0	15.9	14.7	12.6	12.0	13.3	12.7	10.9	9.6
		MCDBR	11.7	16.5	17.3	20.6	22.9	19.8	17.6	16.7	18.7	18.7	11.8	9.9
		MCWBR	10.7	14.2	11.7	11.7	11.3	8.3	7.1	7.5	9.8	11.5	9.3	8.9
	5% WB	MCDBR	11.6	16.2	16.3	19.1	20.7	16.3	14.0	13.6	16.3	17.6	11.6	9.8
		MCWBR	10.7	14.0	11.1	11.0	10.5	7.1	6.2	6.7	8.8	10.9	9.2	8.9
Clear-Sky Solar Irradiance	taub	0.263	0.286	0.322	0.374	0.438	0.430	0.406	0.379	0.334	0.304	0.284	0.266	
	taud	2.222	2.205	2.231	2.222	2.068	2.206	2.332	2.385	2.458	2.468	2.326	2.199	
	Ebn at Noon	770	845	877	865	819	830	846	853	862	829	742	697	
	Edh at Noon	71	95	112	127	156	137	119	108	90	73	64	63	
All-Sky Solar Radiation	RadAvg	1.56	2.68	4.01	4.92	5.64	5.94	5.53	4.76	3.74	2.49	1.50	1.15	
	RadStd	0.07	0.11	0.21	0.34	0.39	0.34	0.39	0.30	0.33	0.17	0.12	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page